

Reasoning Tokens Buy Competence, Not Compliance: What Test-Time Compute Does—and Does Not—Purchase, Measured with a Sign-Exact Oracle

Anonymous authors

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Abstract

Test-time reasoning is usually evaluated on one axis: does more thinking buy more accuracy per token? We add the axis that efficiency work has ignored — *compliance*: when an explicit instruction opposes a convention the model absorbed in training, do reasoning tokens help the instruction win? Using CircuChain v2, a circuit-analysis probe whose answers are verifiable to the sign by four independent solvers, we measure both axes across 18 model configurations from 7 families. Three findings. (1) **Reasoning buys competence only**: toggling the same weights between reasoning and non-reasoning modes at five scales (Qwen3 1.7B–32B) raises magnitude accuracy up to $3\times$ while reversion to the trained convention stays flat — 63–81% (mean 74.2%) everywhere, a band that also contains GPT-5, Gemma-4, GPT-OSS-120B, and DeepSeek-V3 across a $792\times$ competence range. (2) **The GPT-5 reasoning-effort ladder dissociates the axes**: at minimal effort it answers from the prior — cheaply and wrongly; at low effort it solves sign-trap circuits to nine decimals yet still reports the trained frame $\sim 71\%$ of the time; at default effort reasoning consumes the entire token budget and *zero* of 119 requests return an answer at the highest cost per instance. (3) **Unbounded rumination is a real cost**: below the competence threshold, reasoning modes burn the full 16k budget on 73–77% of problems without converging. Across all models, flipping one convention instruction breaks 401 previously-solved problems and fixes 6. Efficiency evaluations that report accuracy-per-token implicitly assume compute purchases controllability; our oracle-verified evidence says it does not. The generator, oracle, and harness will be released; total study cost was under \$100.

1 Introduction

The efficient-reasoning agenda optimizes a ratio: capability delivered per unit of test-time compute (Snell et al., 2024). Its numerator is almost always *accuracy*. This paper measures a second numerator that practitioners implicitly assume comes bundled with the first: *compliance* — whether the model does the task *the way it was told to*, when the instruction opposes a strong prior from training. Instruction-following suites score constraints that do not oppose such priors (Zhou et al., 2023); recent work names the opposed case (Zhang et al., 2025; Wu et al., 2024) but scores it with rubrics or judges rather than an exact oracle.

The distinction matters exactly when models are deployed against specifications: house styles, legacy code conventions, discipline-specific sign disciplines, safety policies that differ from common practice. If reasoning tokens buy accuracy but not obedience, then (i) scaling test-time compute does not close the steerability gap, (ii) efficiency benchmarks that score only correctness will not see the gap at any budget, and (iii) budget tuning (effort levels, early stopping, distillation) can be evaluated on a false frontier.

Measuring this cleanly requires a task where compliance is *objectively decidable at the precision the instruction affects*. We use linear DC circuit analysis: the answer is exact; a sign convention (e.g. the direction mesh currents are defined) changes only the *reporting frame*, never the physics; and the textbook prior is overwhelmingly one-sided (clockwise). CircuChain v2

Scale	No-think		Think (16k budget)	
	Mag. correct	Reversion (n)	Mag. correct	Reversion (n)
1.7b	0.2%	71% (24)	2.4%	72% (69)
4b	0.7%	68% (37)	5.0%	70% (47)
8b	1.2%	70% (61)	2.1%	81% (31)
14b	2.8%	75% (91)	9.3%	78% (91)
32b	5.1%	77% (102)	8.6%	76% (62)

Table 1: The reasoning toggle at fixed weights (Qwen3 ladder). Reversion = share of magnitude-correct sign-diagnostic variables reported in the trained clockwise frame despite the counter-clockwise instruction.

poses 1000 instances — 125 four-way-verified physics configurations $\times$ {default contract, three single-factor convention flips} $\times$ {mesh, nodal} — and grades deterministically against an oracle validated independently of the transform code (125/125 configurations). Within-physics matched pairs give exact McNemar tests: the causal effect of one changed sentence.

Contributions for this workshop. (1) A compute-vs-compliance measurement: the reasoning toggle held at fixed weights across five scales, plus the GPT-5 effort ladder, with per-instance dollar costs. (2) The negative result: no reasoning budget, at any scale or effort, reduces reversion to the trained convention (63–81% everywhere). (3) An efficiency pathology inventory measured with an exact oracle: prior-driven fast-wrong answers at low effort, budget-consuming rumination at high effort, and 100%-truncation regimes where the most expensive configuration returns nothing. (4) A seeded, LLM-free instrument (to be released) cheap enough for budget-sweep studies (full panel $< \$100$).

2 Instrument in brief

CircuChain v2 extends the published CircuChain probe (Ravishankara, 2026). 125 physics configurations across seven linear-DC topologies are accepted only when two independent hand-derived solvers, an exact rational-arithmetic MNA solver, and NGSPICE agree on every quantity; sign traps and controls are balanced $\sim 1:1$ by code predicates. Each physics is posed under the textbook-default contract and three single-factor flips — mesh direction $\text{cw} \rightarrow \text{ccw}$, passive $\rightarrow$ active sign, reference node bottom $\rightarrow$ top — in both mesh (KVL) and nodal (KCL) methods. Expected answers in every frame are validated independently of the transform code, including a *physically re-grounded* SPICE realization of the reference-node flip. Grading is deterministic (structured answer-line parser; magnitude at 2%; sign compliance in the *instructed* frame; truncation recorded as an outcome). Decoding: temperature 0.6, top- p 0.95, 16,384 completion budget for reasoning modes, 4,096 otherwise. Variables whose sign differs between instructed and default frames are *diagnostic*; reversion is measured on those, given correct magnitude.

3 Reasoning raises competence, never compliance

Qwen3’s native /think toggle switches reasoning on/off with weights, prompt, serving stack, and decoding policy fixed — the cleanest possible compute contrast. Table 1 shows the five-scale result: magnitude competence rises with reasoning at every scale (e.g. 14B: 2.8% $\rightarrow$ 9.3%), while reversion to the clockwise frame among magnitude-correct diagnostic variables never moves outside the band (68–81%, all CIs overlapping). Reasoning tokens purchase physics, not obedience.

The cost side. Table 2 shows what the reasoning budget buys at the small end: below the competence threshold, think modes ruminate to the full 16k budget on most problems (truncations: 727–769 of 1000 at 4B–32B) — genuine self-correction spirals that never converge, the overthinking phenomenology (Chen et al., 2024) measured here with an exact oracle.

Scale	No-think		Think	
	Mag. correct	Truncated	Mag. correct	Truncated
1.7b	0.2%	30	2.4%	164
4b	0.7%	1	5.0%	742
8b	1.2%	1	2.1%	732
14b	2.8%	0	9.3%	727
32b	5.1%	4	8.6%	769

Table 2: Rumination economics on the Qwen3 ladder (of 1000 instances). Reasoning modes burn the full completion budget on most instances while magnitude competence stays under 10%.

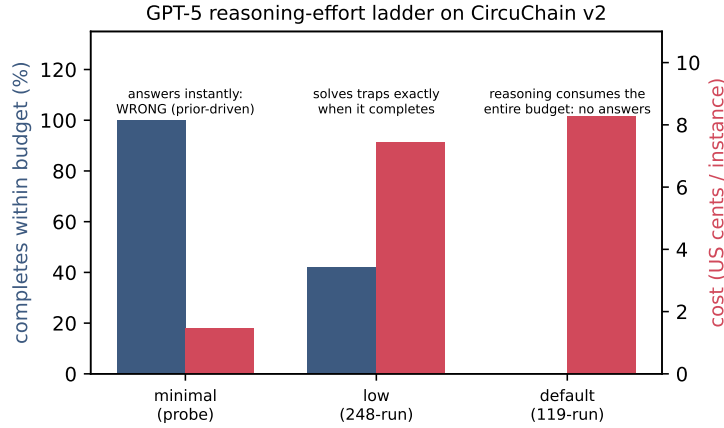


Figure 1: GPT-5 reasoning-effort ladder on CircuChain v2: completion-within-budget rate (left axis) and measured cost per instance (right axis). **minimal** completes always but answers from the prior (wrong); **low** is the only configuration that buys competence; **default** spends the most and returns nothing. Compliance with the counter-normative convention is $\sim 29\%$ regardless.

This is paid-for compute delivering neither competence nor compliance; larger budgets shift the wall rather than remove it.

4 The GPT-5 effort ladder: three budgets, three failure modes

GPT-5’s reasoning-effort control exposes the compute–behavior trade directly on one frontier model (Figure 1):

minimal ($\approx 1.5\text{¢}/\text{instance}$): reasoning tokens = 0; the model answers instantly — *from the prior* — and gets the physics wrong on our probe. **low** ($\approx 7.4\text{¢}$): on a 248-instance run it solves sign-trap circuits exactly when it completes (magnitude-correct 32.7% overall, best of any reasoning configuration we ran under this budget) — and still reports the trained frame on 70.6% of magnitude-correct diagnostic variables; the paired ccw flip breaks 10 problems and fixes 0. **default** ($\approx 8.3\text{¢}$, the most expensive): hidden reasoning consumed the *entire* 8,192-token budget on 119 of 119 requests — zero answers returned, at the highest per-instance cost.

The ladder dissociates the axes clearly: compliance is not purchasable at any of the three budgets, competence is purchasable at exactly one of them, and the default configuration purchases nothing. For efficiency work, the default row is the cautionary tale: without budget-aware evaluation, a deployment can pay maximum cost for a 0% answer rate.

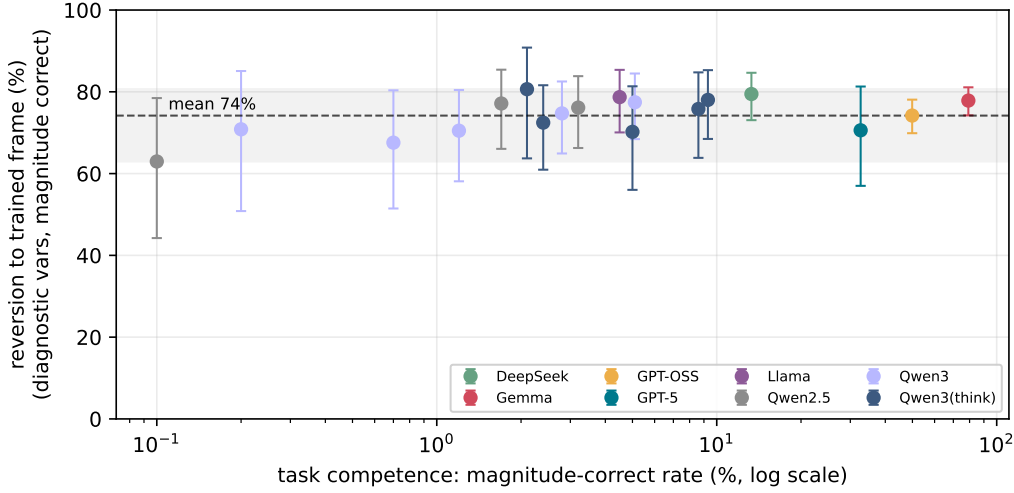


Figure 2: Reversion to the clockwise prior is invariant to competence, scale, family, serving stack, and reasoning mode. Each point is one panel configuration (Wilson 95% CIs; $n \geq 20$ diagnostic variables); shaded band = observed range 63–81%.

5 The invariant that compute never moves

Figure 2 is the study’s central object: reversion to the trained clockwise frame among magnitude-correct diagnostic variables, for every panel configuration with $n \geq 20$, against task competence on a log axis. The band is 63–81% (mean 74.2%) across 18 configurations spanning 7 families, $\sim 792 \times$ in competence, two serving stacks, and both reasoning modes. Aggregated over all models, the within-physics paired flip breaks 401 solved problems and fixes 6 (67:1) — reversion, not noise.

Three controls establish that this is convention-specific reversion rather than generic instruction decay: the same models follow matched *non-convention* unusual instructions (reversed answer order, scientific notation) at 85–97%; GPT-5 and Gemma obey the *other two* convention factors while disobeying mesh direction; and the effect is equal on trap and control physics. The reversion also persists under nodal solving, where loop orientation never enters the derivation — it is a property of the *reporting frame*, not the procedure. Resampling ($k=5$) shows a stable rate (SD 4.7pp) over stochastic items: a biased coin, $\sim 4:1$ toward the trained frame, at every budget.

6 Implications for efficient reasoning

- 1. Put compliance in the efficiency numerator.** Accuracy-per-token frontiers implicitly assume behavior is purchasable with compute. Measured with an exact oracle, it is not: the compliance curve is flat in compute. Efficiency evaluations — especially of effort controls, early-stopping, and distillation — should report *constraint adherence per token* alongside accuracy per token, using probes whose compliance is mechanically decidable (ours will be released; it costs cents per thousand instances).
- 2. Budget-aware evaluation is not optional.** The most expensive GPT-5 configuration returned zero answers; small reasoning models burn full budgets on 73–77% of instances without converging. Truncation must be a first-class reported outcome, or efficiency comparisons silently reward configurations that never finish.
- 3. Effort controls are behavioral, not just economic.** Moving GPT-5 from minimal to low switches it from prior-recitation to genuine derivation — a qualitative regime change

purchased with tokens; more tokens past that point purchase nothing measurable. Efficiency work that tunes effort levels is implicitly selecting *which* failure mode ships.

4. Distillation and steering research have a concrete target. The frame choice is made outside the reasoning trace (it survives nodal solving and every budget), and it is sampled ($\sim 4:1$) rather than committed. A cheap deterministic frame-checking/repair layer recovers more reliability per dollar than any reasoning budget we measured.

7 Limitations

One domain (linear DC circuits); a second convention domain (handedness / cross products) is in progress. Whole-problem effects rest on the few models competent at multi-loop circuits (itself an efficiency datum: 70B-class instruct models sit under 14% magnitude-correct); variable-level results cover the full panel. GPT-5 was run at 248 instances under a bounded budget; its numbers are conservative. Single-sample decoding for most columns is mitigated by the resampling study and the cross-column invariance.

Reproducibility

Seeded byte-reproducible generator (125 physics), four-way oracle, independent transform validation (125/125), deterministic grading, full serving provenance per response, canary-stamped evaluation subset; code and harness to be released upon publication. Total API cost < \$100; local runs on one Apple M5 Max.

References

- M. Ravishankara. CircuChain: Disentangling competence and compliance in LLM circuit analysis. In *Proc. IEEE SoutheastCon*, 2026. arXiv:2602.15037.
- Q. Zhang, X. Lei, R. Miao, et al. Inverse IFEval: Can LLMs unlearn stubborn training conventions to follow real instructions? arXiv:2509.04292, 2025.
- Z. Wu, L. Qiu, A. Ross, E. Akyürek, B. Chen, B. Wang, N. Kim, J. Andreas, Y. Kim. Reasoning or reciting? Exploring the capabilities and limitations of language models through counterfactual tasks. In *NAACL*, 2024.
- J. Zhou, T. Lu, S. Mishra, S. Brahma, S. Basu, Y. Luan, D. Zhou, L. Hou. Instruction-following evaluation for large language models. arXiv:2311.07911, 2023.
- C. Snell, J. Lee, K. Xu, A. Kumar. Scaling LLM test-time compute optimally can be more effective than scaling model parameters. arXiv:2408.03314, 2024.
- X. Chen et al. Do NOT think that much for $2+3=?$ On the overthinking of o1-like LLMs. arXiv:2412.21187, 2024.